Unsupervised detecting anomalies in multivariate time series by Robust Convolutional LSTM Encoder-Decoder

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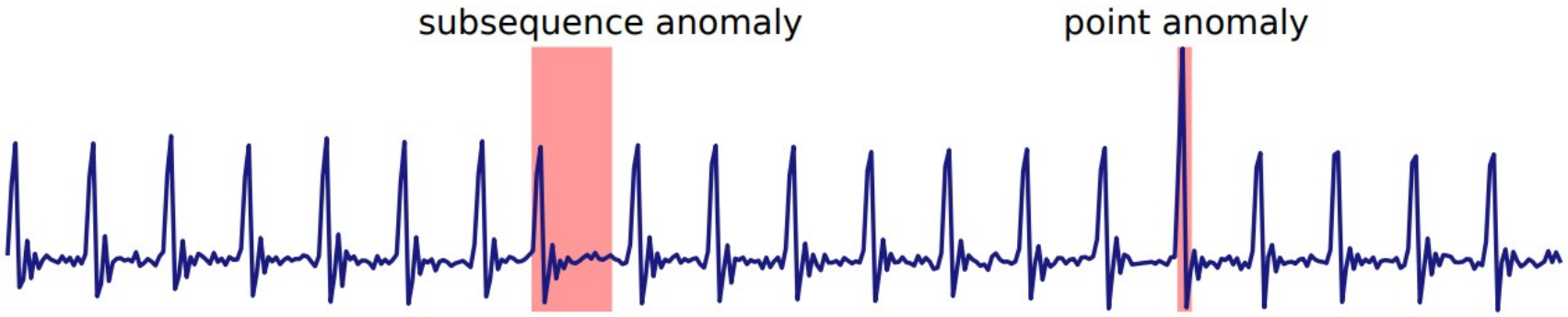
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Introduction

Anomaly Detection (AD) definition (Chandola et al., 2009): Anomaly detection involves identifying patterns in the data that deviate from the expected behavior.

AD for multivariate time series:



Source: Schmidl et al., 2022

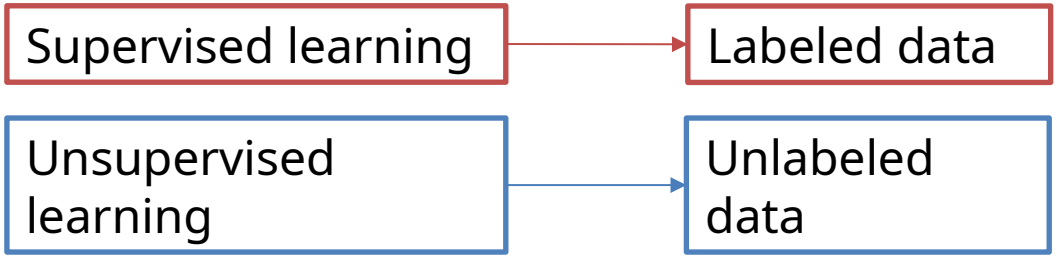
AD applications:



Meaning of the outliers in time series data depending on the aim of the analyst (Blázquez-García et al., 2021)

Unsupervised AD

Supervised vs unsupervised learning (Görnitz et al,. 2013)**:**



Unsupervised AD (Hyun et al,. 2024)**:** In reality, labeling anomalous data points is quite challenging because they occur very rarely and are not similar.

Problem definition (Roth et al,. 2022)**:**

We use $\mathcal{X}_N$ denotes set of nominal time series $(\forall x \in \mathcal{X}_N : y_x = 0)$ available at **training time**, with $y_x \in \{0,1\}$ denoting if point x is **nominal** (0) or **anomalous** (1).

We define $\mathcal{X}_T$ to be the set of sampled provided at **test time**, with $\forall x \in \mathcal{X}_T : y_x \in \{0,1\}$.

$(\mathcal{X} = \{\mathcal{X}^1, \mathcal{X}^2 \ldots, \mathcal{X}^d\}$ can have d univariate time series)
$(\mathcal{X}^d = \{x_1, x_2, \ldots x_T\}$ has the temporal dependence in the timeline $T)$

Unsupervised AD for multivariate time series methods

AD methods based on forecasting/prediction:

Autoregressive model (Abati et al,. 2019):

$$\hat{x}_t = f(x_{t-1}, x_{t-2}, \dots)$$
$$anomaly\ score = g(\hat{x}_t, x_t)$$

Regression model (Hu, Weiming, et al., 2018):

$$x = window[x_{t-1} : x_{t-w}]$$
$$\hat{x}_t = f(x)$$
$$anomaly\ score = g(\hat{x}_t, x_t)$$

AD methods based on reconstruction:

Autoencoder (Yin et al., 2020):

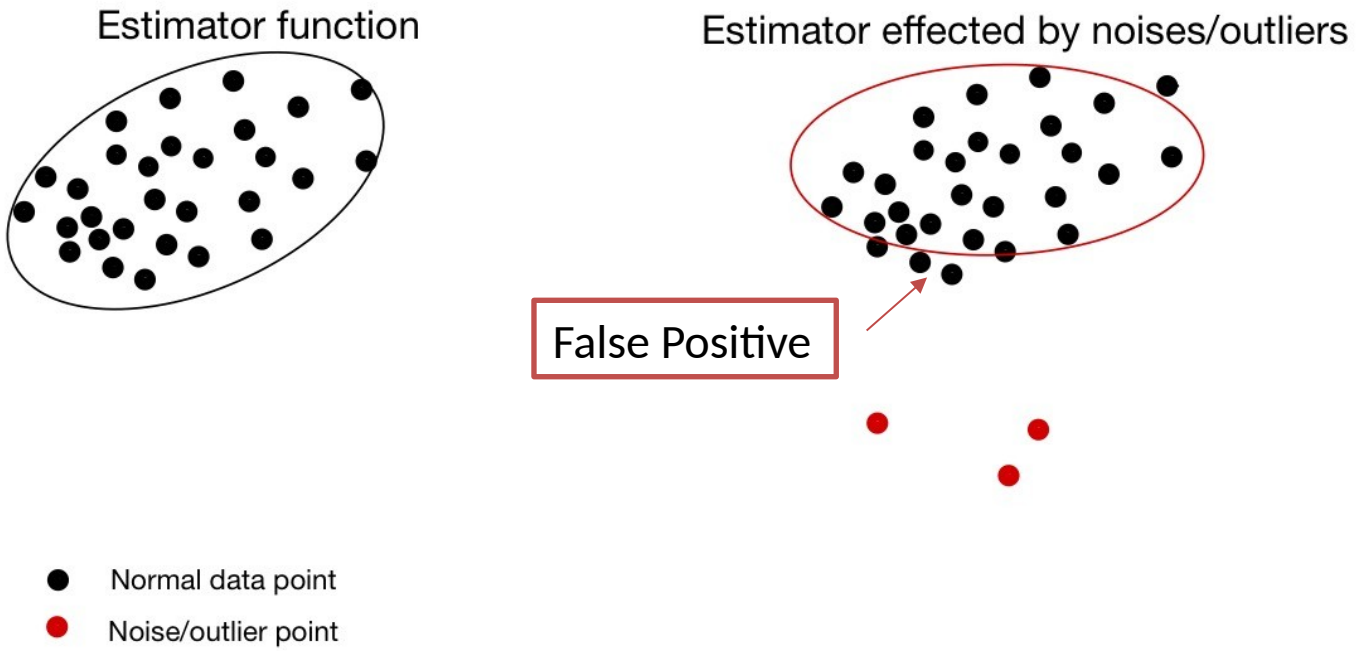
$$x = window[x_{t-1} : x_{t-w}]$$
$$z = f_E(x)$$
$$\hat{x} = f_D(z)$$
$$\begin{cases} anomaly\ score = g(\hat{x}, x) & \textbf{(based on input variable)} \\ anomaly\ score = h(\hat{z}, z) & \textbf{(based on latent variable)} \end{cases}$$

(f: predictor/estimator/encoder/decoder functions)

(g, h: anomaly score functions, i.e. subtraction function)

Challenges and our contributions

Anomaly detection methods are down performance when **training data** has noise.



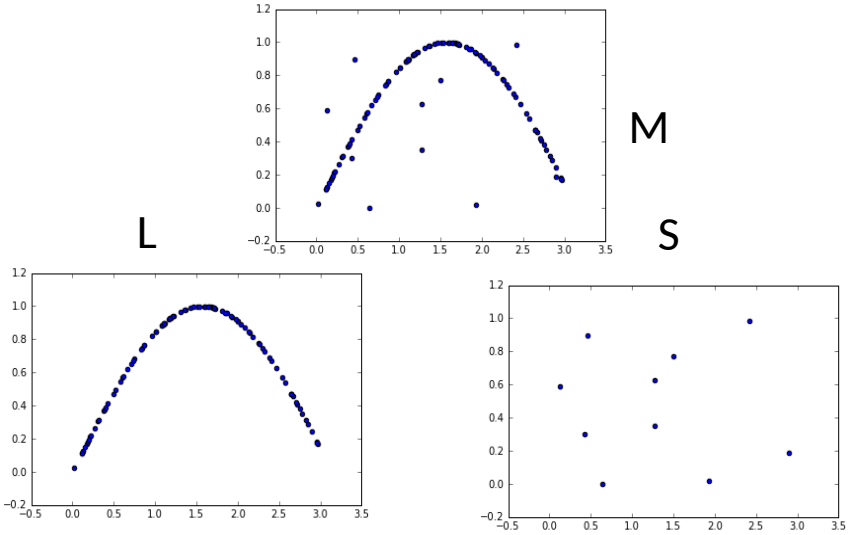
We propose a new training technique that can separate noise/outliers from valuable information in the training dataset

Inspiring Source: Robust Principal Component Analysis

Assumption:

- Assume that the data (M) contains non-relevant information (S) that hinders the generalization of normal data (L).

$$\mathbf{M} = \mathbf{L} + \mathbf{S}$$



Robust Principal Component Analysis (RPCA):

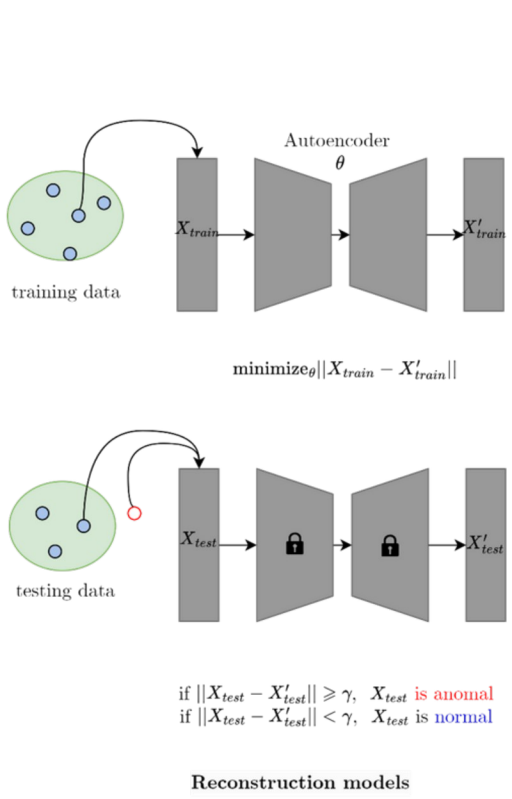
- RPCA can compress normal data and separate noise simultaneously.
- The optimization function of RPCA:

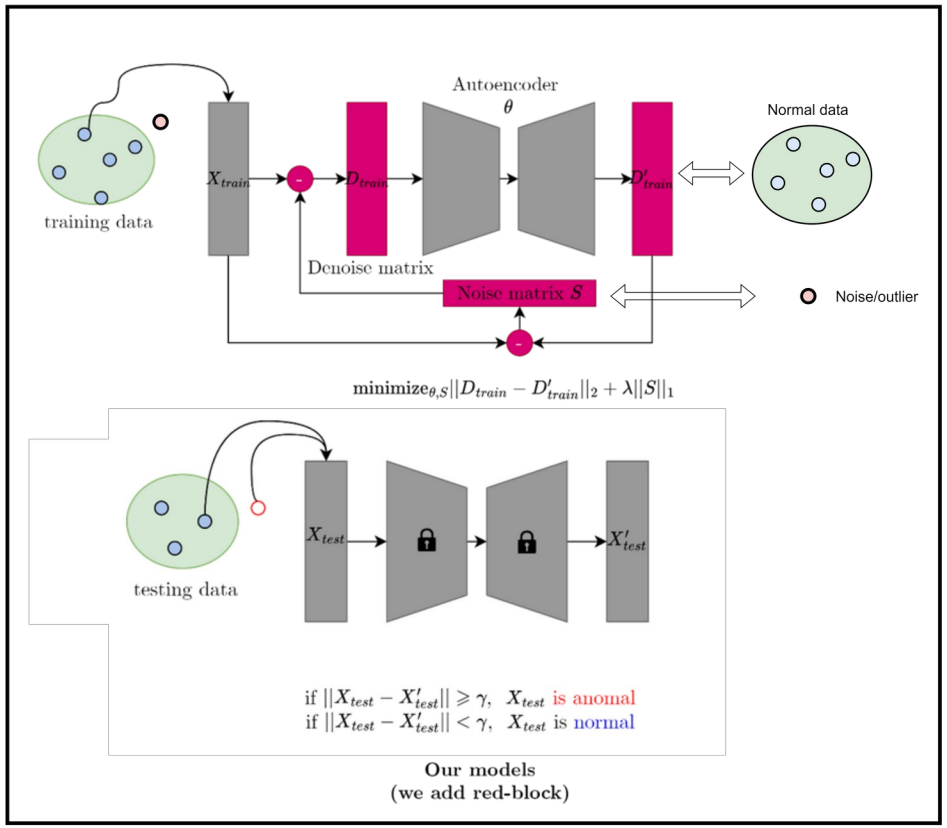
$$\min \rho(\mathbf{L}) + \lambda \|\mathbf{S}\|_0$$
$$\text{s.t.} \|\mathbf{M} - \mathbf{L} - \mathbf{S}\|_F^2 = 0,$$

where ρ(L) is the rank of L, $\|S\|_0$ is the number of non-zero entries in S

Proposed method

We study anomaly detection based on **reconstruction models**
We improve the **robustness** of our models by integrating the Robust Principle Component Analysis (RPCA).

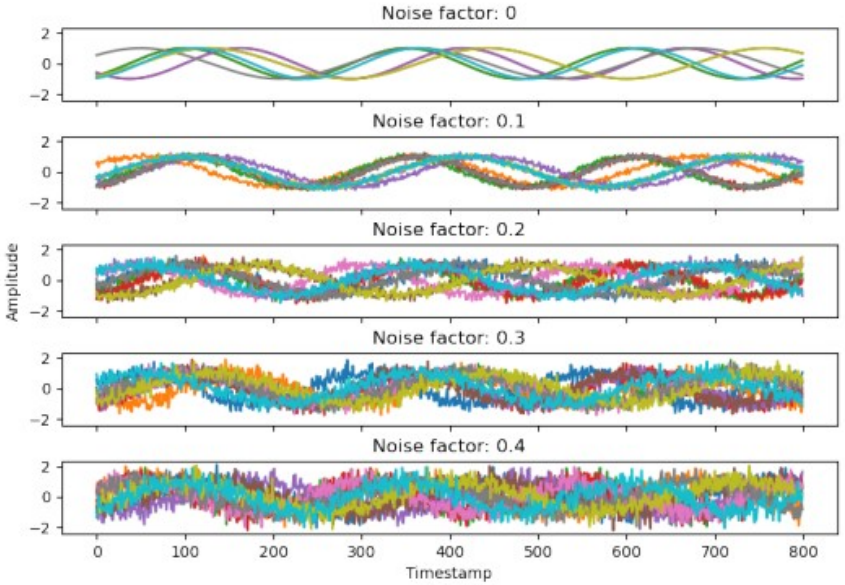




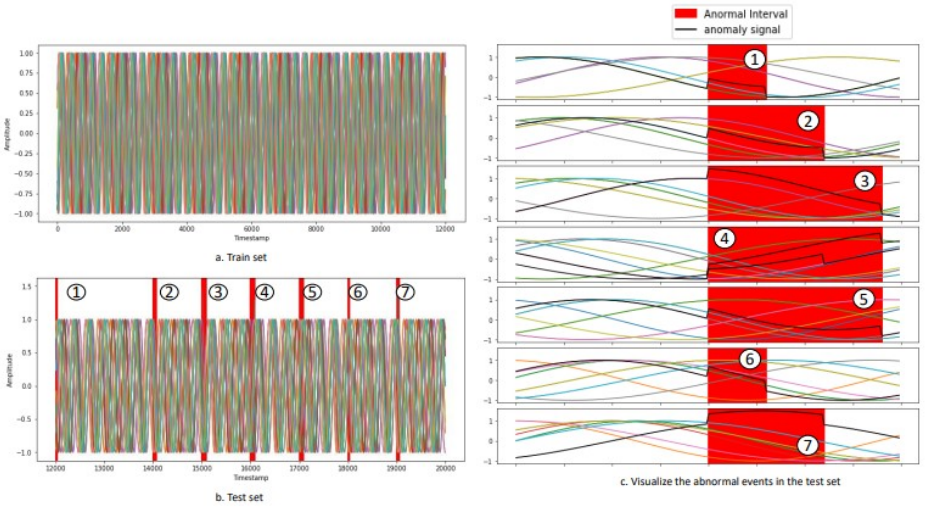
Numerical Experiment: Simulated Datasets

Two datasets: a simulated dataset and a real-world dataset.

- **Simulated data**



Visualize synthetic data with difference noise levels



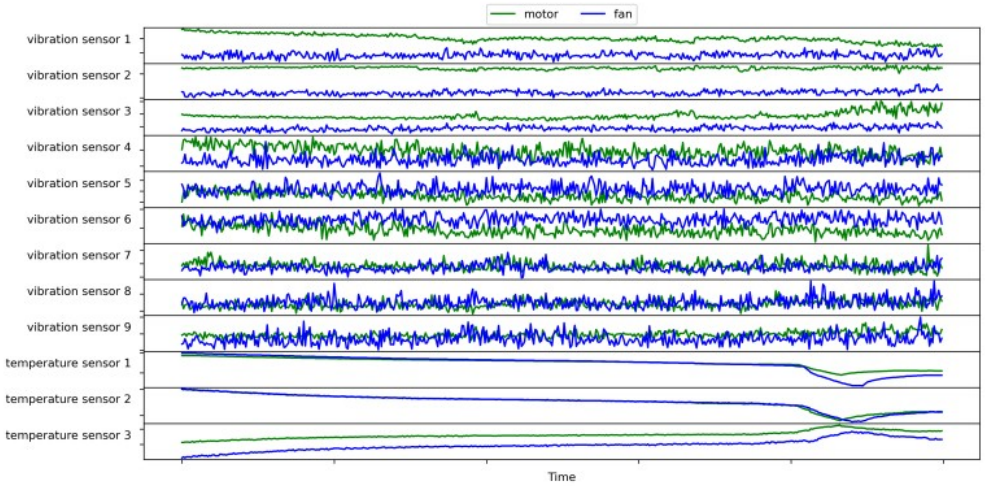
Anomaly events in the test set

Numerical Experiment: KappaX Dataset

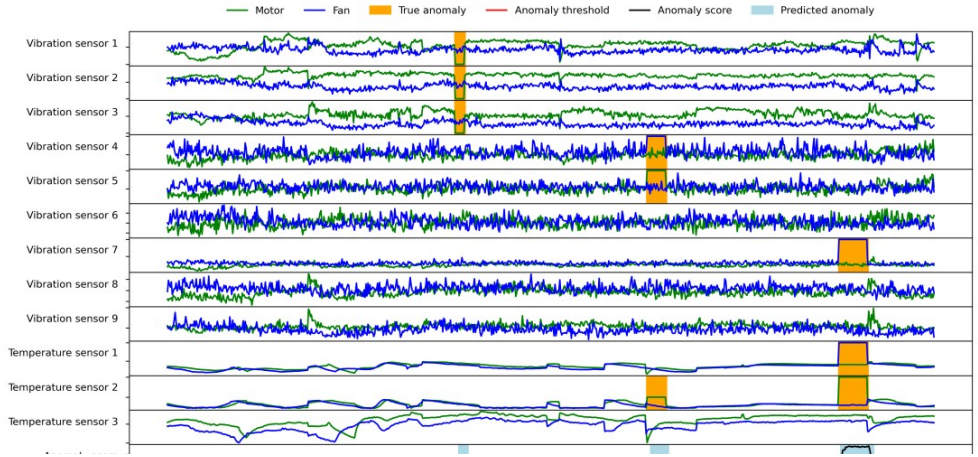
Two datasets: a simulated dataset and a real-world dataset.

- **KappaX dataset provided by Sensoteq**

Real data training set

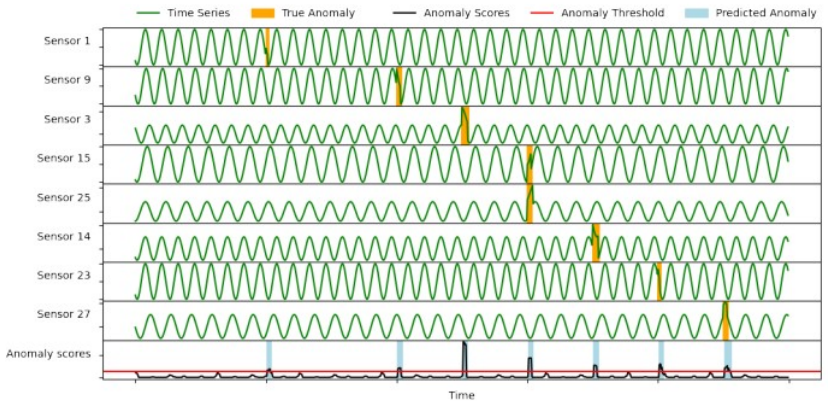


Real data test set

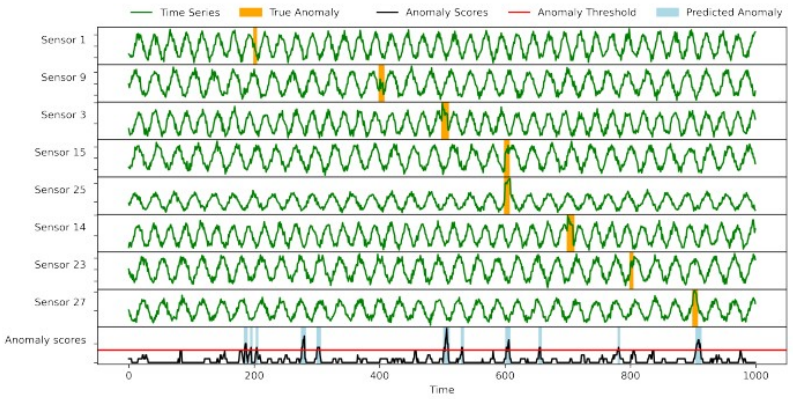


Numerical Experiment: Detecting anomaly

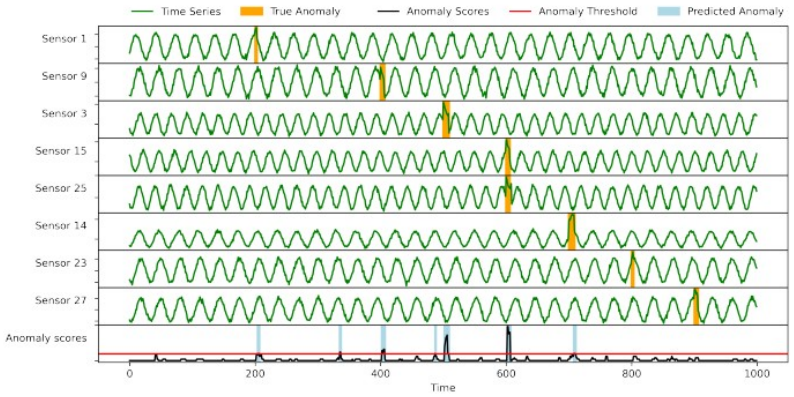
Anomaly detection: Synthetic dataset



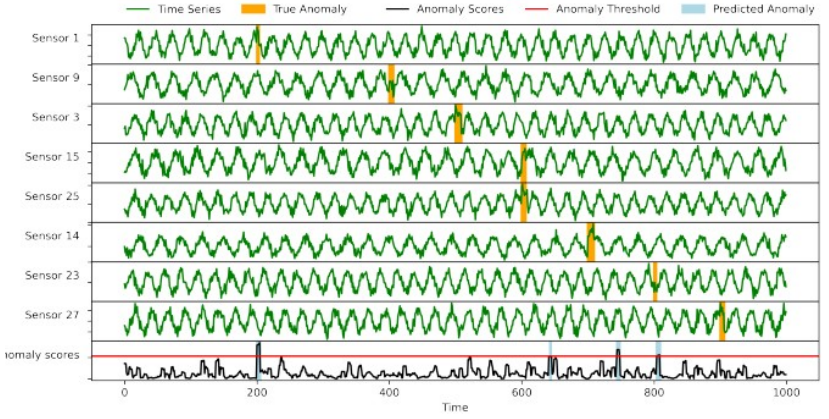
Noise level 0%



Noise level 10%



Noise level 20%



Noise level 30%

Figure 15: Predicted and Ground Truth labels for the Synthetic test set using the RCLED model

Numerical Experiment: Model's performance

RCLED vs. other methods: Synthetic dataset

$$\text{Pre} = \frac{TP}{TP + FP}$$

$$\text{Rec} = \frac{TP}{TP + FN}$$

$$\text{F}_1 = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}}$$

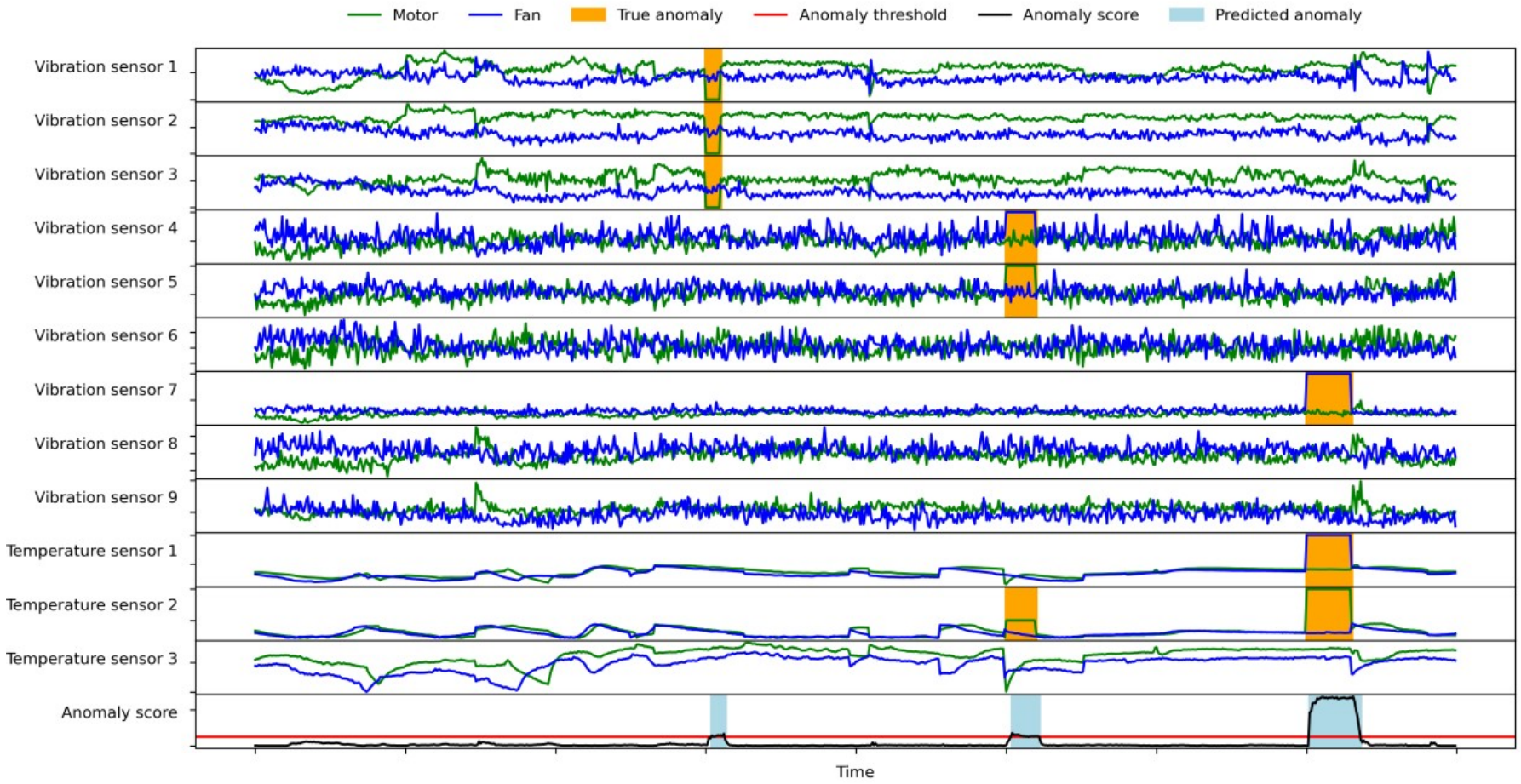
Where:
- TP: true positive
- FP: false positive
- TN: true negative
- FN: false negative

Table 3: The performance of RCLED vs. other methods: Synthetic datasets

Methods	Noise level: 0%			Noise level: 10%		
	Pre	Rec	F1	Pre	Rec	F1
OCSVM-Linear	0.32	0.52	0.40	0.14	0.18	0.15
OCSVM-RBF	1.00	0.44	0.62	0.27	0.12	0.17
HBSO	0.44	0.06	0.11	0.1	0.02	0.03
GMM	0.33	0.25	0.25	0.25	0.10	0.17
DeepLog	0.47	1.00	0.64	0.33	0.15	0.21
DGMM	0.33	0.20	0.25	0.33	0.20	0.25
LSTM	0.48	0.85	0.61	0.32	0.29	0.30
LSTM-ED	0.43	0.36	0.39	0.33	0.12	0.18
MSCRED	0.38	0.24	0.30	0.67	0.13	0.22
Ours (RCLED)	**0.80**	**0.55**	**0.65**	**0.68**	**0.4**	**0.51**
Gain (%)	41.3	-	1.5	51.5	50.0	50.9

Methods	Noise level: 20%			Noise level: 30%		
	Pre	Rec	F1	Pre	Rec	F1
OCSVM-Linear	0.07	0.09	0.08	0.05	0.06	0.06
OCSVM-RBF	0.10	0.04	0.06	0.05	0.02	0.03
HBSO	0.25	0.02	0.04	0.00	0.00	0.00
GMM	0.00	0.00	0.00	0.00	0.00	0.00
DeepLog	0.08	0.20	0.11	0.04	0.00	0.01
DGMM	0.08	0.19	0.11	0.09	0.02	0.04
LSTM	0.09	0.16	0.12	0.05	0.03	0.04
LSTM-ED	0.05	0.71	0.09	0.25	0.02	0.04
MSCRED	0.00	0.00	0.00	0.00	0.00	0.00
Ours (RCLED)	**0.31**	**0.39**	**0.35**	**0.28**	**0.24**	**0.26**
Gain (%)	70.9	58.9	65.7	71.4	75.0	76.9

Numerical Experiment: Detecting anomaly



Predicted and Ground Truth labels for the KappaX dataset provided by Sensoteq, Ireland

Numerical Experiment: Model's performance

We also compare our models' performance (RCLED) with other models.

Models	KappaX		
	Pre	Rec	F1
OC-SVM-Linear	0.32	0.95	0.48
OC-SVM-RBF	0.24	1.00	0.39
HBSO	0.44	0.06	0.11
GMM	0.48	0.83	0.61
DeepLog	0.47	1.00	0.64
DGMM	0.33	0.20	0.25
LSTM-NDT	1.00	0.56	0.72
LSTM-ED	1.00	0.61	0.76
MSCRED	1.00	0.70	0.82
RCLED	**0.83**	**0.87**	**0.85**
Gain (%)	-	19.5	3.5

Conclusion

- We are introduced to the **RCLED (Robust Convolutional Long Short-Term Memory Encoder-Decoder).**

- The RCLED model is designed with an encoder-decoder architecture **and has the ability to work with noisy and multivariate time series data**.

- Compared to other methods such as OCSVM, GMM, HBSO, GMM, DGMM, DeepLog, LSTM, LSTM-ED, and MSCRED, **RCLED** is a robust autoencoder approach that enhances the modeling capabilities of autoencoders by enabling them to **handle noise** in **multivariate time series** effectively.

- We also introduce the **KappaX dataset**, a real-world dataset collected from industrial production systems. **RCLED has been tested on this dataset** and **demonstrated an accurate anomaly detection capabilities model** for anomaly detection in multivariate time series.

- Finally**, the RCLED model is a powerful approach for anomaly detection in multivariate time series**. With its ability to handle noisy data and its robustness to unpredictable patterns, **RCLED shows great promise in real-world applications**. It outperforms other methods and provides more accurate predictions, making it a valuable tool for anomaly detection systems where false positives can have significant consequences.

References

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Thank for listening